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Enhanced short-term traffic flow forecasting using a meteorological spatio-temporal transformer with multi-head attention and whale optimization

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Rapid urbanization and increasing vehicle ownership have intensified traffic congestion, creating serious social, economic, and environmental challenges. Accurate short-term traffic flow prediction is therefore essential for effective urban traffic management and congestion mitigation. While traditional models struggle with the non-linear, dynamic nature of traffic data, our approach proposes an optimized synthesis named the Meteorological Spatio-Temporal Transformer Network with Multi-Head Attention and Whale Optimization Algorithm (MSTT-MHA-WOA). This framework enhances prediction by uniquely integrating non-linear meteorological features and, critically, by using the Whale Optimization Algorithm (WOA) for autonomous hyper parameter tuning. We evaluate the model's robust and consistent performance across six diverse, real-world traffic scenarios–a crucial test of its generalization capability under complex conditions.

Keywords Short-term traffic flow forecasting, Spatio-temporal dependencies, Transformer network, Whale optimization algorithm (WOA), Meteorological features integration, Hyperparameter tuning, Traffic congestion mitigation

Rapid urbanization and increasing vehicle ownership have intensified traffic congestion, creating serious social, economic, and environmental challenges. Accurate short-term traffic flow prediction is therefore essential for effective urban traffic management and congestion mitigation, enabling authorities to deploy dynamic tactics like rerouting and signal timing adjustments. Traffic prediction is inherently challenging due to the non-linear, dynamic nature of the data, which is heavily influenced by factors such as time of day, road conditions, and weather changes.

Early models, such as ARIMA and its variations, primarily focused on the temporal aspect but struggled to address the intricate spatial relationships in road networks. Although these statistical methods performed well in controlled settings, they often failed in real-world scenarios where external elements like weather significantly affect traffic trends. In response, recent machine learning models, including those integrating Long Short-Term Memory (LSTM) (e.g., EnLSTM-WPEO[1]) and Graph Convolutional Networks (GCN) (e.g., ASTGCN[2]), have improved accuracy by modeling both spatial and temporal dependencies. The integration of Transformer-based models is among the most promising strategies, given their capacity to recognize extended dependencies. Models like ASTTN[3] and DSTAGNN[4] specifically address complex cross-spatial–temporal correlations, resulting in improved accuracy.

Despite these efforts, significant limitations remain in current State-of-the-Art (SOTA) models like MTESformer[5] and AGC-LSTM[6]. For instance, MTESformeromits external variables like weather, which are vital for real-world predictions. Similarly, AGC-LSTM suffers from reduced accuracy in coarser time intervals and requires exhaustive retraining for different sectors, limiting its generalization capability. While Spatio-Temporal networks are standard, a critical knowledge gap remains in developing a single, robust model that can simultaneously integrate highly non-linear external factors (such as adverse weather) and utilize automated

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optimization to guarantee stable, high accuracy when faced with the vastly different dependency patterns inherent in diverse traffic conditions.

Our work addresses this challenge by introducing the Meteorological Spatio-Temporal Transformer Network with Multi-Head Attention and Whale Optimization Algorithm (MSTT-MHA-WOA). This optimized and integrated framework enhances short-term traffic prediction by uniquely incorporating meteorological data and leveraging the Whale Optimization Algorithm (WOA) to autonomously fine-tune the model's configuration. We evaluate the model's superior generalization and robustness across six diverse traffic scenarios, demonstrating its ability to outperform current SOTA methods under complex and fluctuating conditions.

The primary methodological contribution of this work, relative to existing Transformer-based models, is the development of a unified framework that fuses non-linear meteorological variables (temperature and precipitation) directly into the spatio-temporal attention mechanism. Unlike standard models that require manual hyperparameter tuning, our approach utilizes the Whale Optimization Algorithm (WOA) to automatically adapt the model's architecture, ensuring high predictive stability across diverse conditions such as holiday peaks and weekend traffic.

The primary contributions of this work are summarized as follows:

- **Multi-source data fusion:** A framework is proposed that integrates high-resolution meteorological data (precipitation, temperature, and visibility) with historical traffic patterns to enhance forecasting accuracy under adverse conditions.
- **Enhanced architecture:** A Multi-Source Traffic Transformer (MSTT-MHA) is developed, utilizing Multi-Head Attention to capture complex spatial–temporal dependencies.
- **Automated optimization:** A Whale Optimization Algorithm (WOA) is implemented to automate the selection of optimal hyperparameters, thereby eliminating manual trial-and-error and ensuring model stability.

The remaining sections of this paper are organized as follows:

- Section "Related works" reviews the literature, assessing recent advancements in spatio-temporal and deep learning techniques for traffic flow prediction.
- Section "Data source" outlines the dataset used in this research, detailing the preprocessing steps taken to incorporate both traffic and meteorological data.
- Section "Methodology" explains the proposed methodology, detailing the design of the Spatio-Temporal Transformer Network, the Multi-Head Attention mechanism, and the Whale Optimization Algorithm used for model tuning.
- Section "Experimental setup & evaluation metrics" outlines the experimental setup and procedures.
- Section "Results & discussions" details the experimental findings, evaluates the proposed model against the current state-of-the-art approaches, and provides an in-depth discussion of the results.
- Section "Conclusion" concludes the paper and recommends potential lines of inquiry for further study.

Related works

Short-term traffic flow prediction

Short-term traffic flow prediction is crucial for effective traffic management due to the system's nonlinear and dynamic nature. To improve prediction accuracy, models such as SVR with Bayesian Optimization have been proposed, enhancing performance by iteratively fine-tuning parameters[7]. This motivates our adoption of adaptive learning techniques to optimize prediction outcomes. Similarly, the T-LSTM model captures daily temporal correlations and effectively handles missing data[8], further supporting our focus on robust temporal modeling under incomplete or noisy traffic conditions.

A hybrid deep learning model combining AutoEncoder and GRU captures spatial–temporal correlations by utilizing upstream and downstream link relationships, thereby improving prediction accuracy[9]. Building on this idea, the importance of joint spatial–temporal modeling becomes evident and forms a key part of the proposed approach. Extending further, Convolutional LSTM networks predict multi-lane traffic by considering spatial–temporal dependencies, achieving high accuracy on the US I-101 dataset[10], which reinforces the effectiveness of integrated spatial–temporal learning adopted in our model. Moreover, the STSGCN model pushes this advancement by synchronously modeling space–time data to capture key correlations and heterogeneities[11]. Inspired by these developments, the proposed method enhances spatial–temporal learning through advanced graph-based mechanisms.

Traffic flow prediction benefits from combining spatial and time-based trends. A Spatio-temporal neural graph model with positional attention captures dependencies[12], a concept that aligns with the attention mechanism used in the proposed model to handle spatial–temporal dependencies. Similarly, a CNN-BiGRU-attention model has shown success in forecasting[13], further supporting the proposed approach that integrates multi-feature fusion. Additionally, a multi-head attention transformer outperforms GRU and LSTM in capturing distant dependencies[14], paralleling the proposed model's use of transformers to capture long-range dependencies. Furthermore, SARIMA-NAR models achieve high accuracy in nonlinear environments[15], providing a foundation for the proposed approach's focus on improving seasonal forecasting.

A deep learning model (GCST-GRU) captures spatio-temporal dependencies, outperforming others[16], a feature that mirrors the proposed graph-based approach. Similarly, the DGCRIN model enhances data imputation for incomplete datasets[17], which relates directly to the proposed model's method for handling missing data. The DGCRN model, utilizing hyper-networks, excels in attribute extraction[18], paralleling the feature extraction technique proposed in our model. STRGCN combines residual learning and attention mechanisms for improved prediction accuracy[19], aligning closely with the attention-driven optimization used in the proposed approach.

Additionally, MGCN-WOALSTM enhances predictions by incorporating weather data[20], a concept that is also applied in the proposed model.

A review of LSTM-based models for short-term traffic estimation[21] highlights their strengths and weaknesses, providing valuable insights for selecting the most suitable models under varying traffic conditions. Similarly, models like STGCN[22] focus on traffic-specific data modeling, which directly relates to the data enhancement techniques employed in the proposed method. For long-term traffic prediction, MTESformer[6] effectively captures complex spatio-temporal dependencies, aligning with the approach adopted by the proposed model. Additionally, a dual-stage framework utilizing surveillance camera data has been shown to improve data robustness and deliver better short-term prediction performance compared to traditional models.

In air traffic management, the AGC-LSTM model captures spatial relationships among routes, reducing MAE by 14.4% compared to traditional models[5], which aligns with the proposed approach's method of handling spatial dependencies. Similarly, a phase-based prediction method for isolated intersections employs multi-task learning to incorporate external factors, improving accuracy across different time scales[23], mirroring the proposed method's approach to handling external factors. Additionally, the MLR-LSTM model addresses challenges posed by incomplete data, enhancing robustness during data loss[24], which relates to the proposed method's resilience in dealing with missing data.

The Levenberg–Marquardt neural network predicts non-autonomous vehicle traffic with high accuracy[25], aligning with the proposed model's emphasis on accuracy. Similarly, a model that balances efficiency and accuracy in large-scale forecasting uses autoregressive models[26], which is akin to the proposed approach. Attention-driven models enhance dynamic forecasting by capturing dependencies[27], closely paralleling the proposed method's use of spatiotemporal dependencies. A hybrid LSTM and Graph Convolutional Network improves predictions in data-limited environments[28], which is relevant to the proposed approach's ability to handle similar challenges. Additionally, a spatiotemporal network that incorporates weather conditions provides better stability[29], mirroring the proposed model's integration of external factors. Lastly, combining wavelet-based denoising with LSTM and GRU achieves high forecasting accuracy[30], similar to the proposed model's focus on capturing time trends.

The MVSTT was introduced by[31], to improve traffic-flow forecasting. It captures short- and long-term patterns and handles fixed and dynamic spatial dependencies. The proposed MSTT-MHA-WOA model builds on this idea by incorporating multi-head attention and optimization to further improve prediction accuracy. The ProSTformer presented by[32], a progressive space–time self-attention model designed for efficient and accurate short-term traffic flow forecasting. By factorizing spatial and temporal dependencies and applying attention progressively from local to global contexts, the model improves focus on relevant patterns. Techniques like Tensor Rearranging and Patch Merging reduce computational overhead while preserving essential dependencies. This approach highlights recent advances in pure attention-based models and offers an effective solution for handling large-scale traffic data.

Spatio-temporal feature learning

Spatio-temporal feature learning plays a crucial role in modeling traffic flow, given the complex interactions between time and location. GMAN[33] employs attention modules to capture these dependencies and enhance prediction accuracy, aligning with the spatio-temporal attention strategy adopted in our work. ASTGCN[2] leverages attention-based mechanisms to incorporate current, daily, and weekly patterns, a concept that resonates with the dynamic temporal modeling in our method. A multi-objective learning model optimized through particle swarm techniques[34] effectively addresses spatio-temporal heterogeneity, a challenge also considered in our optimization approach. Similarly, T-GCN[35] integrates GCNs with GRUs to simultaneously learn spatial and temporal dynamics, a structure closely mirrored in our proposed framework.

Hybrid models have demonstrated significant effectiveness in traffic prediction tasks. For instance, the LSTM-VMD framework[36] achieved accurate multistep forecasts by decomposing traffic patterns, a strategy similar to our modular approach for capturing temporal features. Similarly, the deployment of an LM neural network in Rome[37] supported urban traffic planning, highlighting the real-world applicability of our model. Furthermore, ASTGNN[38] integrated self-attention with dynamic graph convolutions, aligning with our method in addressing both spatial heterogeneity and periodic trends.

Graph-based learning methods have seen significant advancements in recent years. DSTAGNN[4] improves spatio-temporal modeling by leveraging multi-head attention and gated convolutions, which aligns with our objective of capturing dynamic spatio-temporal interactions to enhance prediction accuracy. Similarly, ASTTN[3] integrates self-attention with adaptive graphs for robust traffic prediction, further supporting our focus on transformer-based architectures for effective spatio-temporal correlation learning.

Knowledge representation within spatio-temporal graph convolutional networks has also been actively explored. A graph-based knowledge representation approach[39] enhances traffic forecasting by improving traditional models, supporting our goal of strengthening prediction through structured spatio-temporal learning. Similarly, the HiSTGNN model[40] captures regional interactions and long-term temporal patterns through adaptive graph learning, aligning with our focus on uncovering hidden spatial dependencies to boost forecasting accuracy.

Urban traffic forecasting remains a complex challenge due to the dynamic nature of road structures. A model combining GCNs, GRUs, and multi-head attention[41] successfully captured key spatio-temporal shifts, achieving 91.8% accuracy on the LA dataset, which supports our objective of modeling evolving traffic patterns. Similarly, a graph-based method[42] enhanced real-time forecasting by effectively modeling complex road networks, aligning with our focus on structural traffic data. Furthermore, a Bi-GRU model with multi-range mask graph convolution[43] learned recent and periodic trends while filtering out noise, complementing our aim of efficiently extracting meaningful spatio-temporal features.

The recent STGAGRTN model by[44] addresses major challenges in traffic flow forecasting by effectively capturing dynamic spatial–temporal dependencies and optimizing computational complexity. It extracts long-term temporal patterns and spatial relationships beyond neighboring nodes, offering improved prediction accuracy. This approach highlights recent advancements and provides a strong reference point for exploring further improvements in forecasting models.

PSPJSTGCN model was proposed by[45], which integrates both self-learned and pre-defined spatial graphs in parallel to enhance traffic flow forecasting. By combining prior knowledge with data-driven structures, the model captures richer spatial–temporal patterns. It also employs multi-scale gated convolution to effectively extract dynamic temporal dependencies. This recent work underscores the trend of hybrid graph structures and provides a strong benchmark for developing more adaptive and accurate forecasting models.

A hierarchical model combining spatio–temporal graph convolutional networks and a transformer proposed by[46] to improve traffic flow forecasting. Their model captures both short-term and long-term dependencies and solves the over-smoothing issue in GCNs using an attention fusion mechanism. This approach is relevant to the proposed research as it highlights an effective way to integrate spatial and temporal features, which is essential for accurate prediction in complex dynamic systems.

A DAGNBL framework proposed by[47] combining BiLSTM and GNN to forecast traffic flow, incorporating weather and accident data. The model uses attention modules to identify important spatial and temporal dependencies, achieving high prediction accuracy. This approach is relevant to the proposed research as it demonstrates the effectiveness of hybrid deep learning models in integrating multiple factors for improved traffic forecasting.

The function of deep learning in traffic prediction

Deep learning models have advanced traffic prediction by learning complex patterns from large datasets.[48] Introduced the DNN-BTF model, which used attention mechanisms to capture daily and weekly periodicity, leveraging CNNs for spatial features and RNNs for temporal features. This approach aligns with the proposed work's integration of CNNs and RNNs for improved spatio-temporal forecasting. In another approach,[1] developed the EnLSTM-WPEO model, combining deep learning techniques with optimization methods for short-term traffic forecasting. This approach aligns with the proposed work's integration of deep learning and optimization for improved prediction accuracy. Building on the Graph Multi-Attention Network (GMAN) introduced by[33], the architecture was refined for long-term prediction by integrating multiple spatio-temporal attention blocks, mitigating error propagation and improving accuracy. This connects with the proposed work's focus on enhancing model architectures for better long-term traffic forecasting.

A hybrid framework combining a Stacked Autoencoder with a Radial Basis Function Neural Network, proposed by[49], captured temporal correlations and the effect of climatic conditions on traffic dynamics. This relates to the proposed work, as it also focuses on incorporating external factors to enhance prediction accuracy. The CDLP model, described by[50], utilized CNN-LSTM-attention and CNN-GRU-attention frameworks to capture both local trends and long-term dependencies, outperforming baseline models. The proposed work similarly integrates advanced attention mechanisms to improve prediction accuracy and handle complex dependencies in traffic flow.

STTNs introduced by[51] improved long-term traffic forecasting by modeling spatial and temporal relationships. This connects to the proposed work, which also aims to enhance long-term predictions with dynamic modeling techniques. Lightweight Transformer Networks, proposed by[52], addressed computational complexities and enhanced prediction accuracy for time series. Similarly, the proposed work leverages modified transformer architectures to balance prediction accuracy and efficiency for traffic flow forecasting.

Hybrid models combining LSTM, GRUs, and other deep learning architectures have shown promise in improving traffic flow prediction accuracy by better capturing complex spatio-temporal patterns[53]. Similarly,[54] demonstrated a spatio-temporal graph convolutional network integrating weather factors to enhance long-term predictions, aligning with the proposed model's use of external attributes. Finally,[55] introduced SATP-GAN for time-series prediction, which complements the proposed model's exploration of generative approaches for traffic forecasting. This supports the proposed work by highlighting predictive improvements across multiple junctions with data preprocessing.

A GAN-based model employing self-attention and reinforcement learning achieved a 6.5% improvement in traffic time-series prediction over baseline methods, it demonstrating the potential of generative approaches in this domain. Meanwhile, deep models combining CNNs for spatial and LSTMs for temporal data, like the CLTFP model[25], have shown enhanced vehicle flow prediction, supporting our focus on hybrid spatial–temporal architectures. The ASTTGN model was introduce by[56], which effectively captures complex spatial and temporal dependencies in traffic data using adaptive transformer modules. By combining dynamic spatial and temporal attention with gated feature aggregation, the model improves forecasting accuracy from IoT loop detector data. This work emphasizes the importance of adaptive mechanisms and attention-based architectures in handling real-world spatial–temporal traffic patterns.

The AGCRNproposed by[57], which enhances traffic forecasting by using two adaptive modules: NAPL for node-specific patterns and DAGG for automatic dependency inference, without relying on pre-defined graphs. This approach is relevant to the proposed research as it highlights adaptive learning techniques for more accurate traffic prediction without fixed spatial structures. The literature on traffic flow prediction highlights the success of spatio-temporal models using attention mechanisms, graph convolutions, and hybrid deep learning. Models like MVSTT and ProSTformer improve accuracy by capturing dynamic spatial and temporal dependencies, while GMAN and ASTGCN emphasize attention and graph learning. However, handling missing data, optimizing computational complexity, and maintaining model efficiency in large-scale environments remain critical challenges that need to be addressed. The proposed work aims to address these disadvantages by integrating

multi-head attention, graph-based learning, and external factors to enhance prediction accuracy, robustness, and efficiency.

Data source

Study area

The experiment utilizes a combination of traffic and weather data from the year 2023, specifically focusing on State Route 94 West at CA PM 7.24 (Abs PM 5.8) in San Diego. This location, situated near residential, commercial, and recreational zones, often experiences heavy traffic, making it ideal for urban congestion analysis. Traffic flow data was collected along College Grove Way toward the westbound direction of Highway 94 (94 WB). The traffic data, obtained from the PeMS website, includes features such as Date, Vehicle Miles Traveled (VMT), Lane Points, and % Observed Traffic.

Complementary weather data was sourced from the NOAA (National Oceanic and Atmospheric Administration) Local Climatological Data (LCD), which provides hourly atmospheric details from nearby weather stations. This data will support predictive models to optimize traffic flow and inform evidence-based transportation policies, helping to alleviate congestion in this critical corridor. The raw meteorological dataset for this study contained 87 distinct features. To identify the most impactful variables for traffic flow prediction and avoid model overfitting from irrelevant data, a preliminary Feature Correlation Analysis was conducted. This analysis, utilizing both Pearson and Spearman correlation coefficients, indicated that Temperature (T) and Precipitation (P) showed the highest and most consistent non-linear correlation with traffic volume. Their direct and significant impact, particularly during adverse conditions, made them the optimal choice for integration into our model. Consequently, only these two features were selected and vectorized to represent the external environmental conditions. This focused approach ensures the model prioritizes the most relevant information, thereby enhancing both its accuracy and computational efficiency (Figs. 1 and 2).

In this study, six distinct traffic flow scenarios have been analyzed to understand their dynamics during peak hours. The scenarios include weekday traffic, weekend traffic, and holiday traffic, each evaluated during specific time periods. The morning peak is defined as the interval from 7:00 AM to 9:00 AM, while the evening rush spans from 4:00 to 8:00 PM. Table 1 provides a summary of the scenarios and the corresponding time periods.

Integrated traffic and meteorological data sources

The dataset employed in this research is integrated from two primary sources: the PeMS (Performance Measurement System) and the National Weather Service (NWS). The PeMS data provides crucial traffic flow information, including 5-min interval traffic counts across key transportation corridors in San Diego County. Meanwhile, the NWS contributes detailed weather parameters such as temperature and precipitation. Together, these sources offer a rich, time-specific (from January 1, 2023, to May 31, 2023) dataset, enabling a comprehensive analysis of traffic and weather patterns in the study area.

The dataset construction follows a continuous time-series approach for the training phase to ensure the model captures long-term temporal dependencies. For evaluation, the data is categorized into six distinct scenarios: weekday morning/evening peaks, weekend morning/evening peaks, and holiday morning/evening peaks. This allows for a granular assessment of how the model performs during specific high-demand periods without fragmenting the initial learning process of the Transformer.

Data preprocessing

Preprocessing data is essential for refining the integrated dataset sourced from Caltrans PeMS and NOAA. Various techniques such as data cleaning, normalization, outlier detection, and temporal alignment were applied

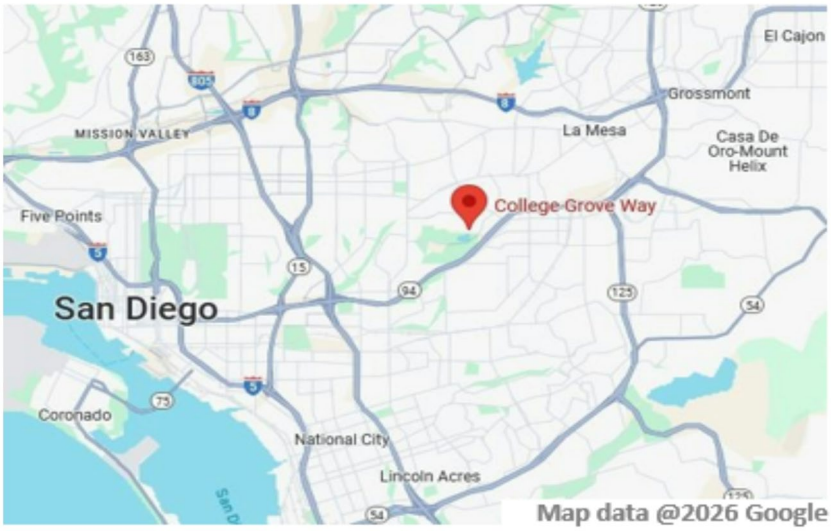


Fig. 1. Geographic overview of the study area within District 11, San Diego County. Map data ©2026 Google.

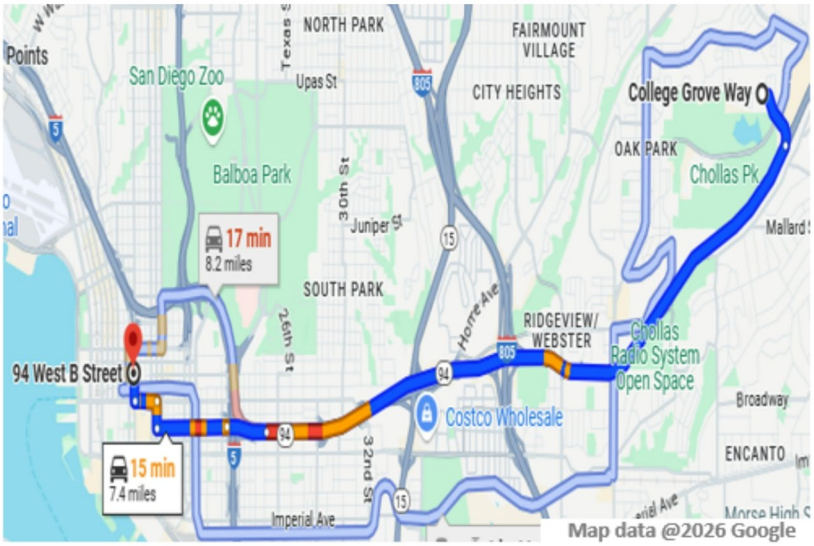


Fig. 2. Detailed road network from College Grove Way to 94 WB, San Diego. Map data ©2026 Google.

Scenario description	Time period	Peak type
Weekday traffic	7:00 AM–9:00 AM	Morning peak
Weekday traffic	4:00 PM–8:00 PM	Evening peak
Weekend traffic	7:00 AM–9:00 AM	Morning peak
Weekend traffic	4:00 PM–8:00 PM	Evening peak
Holiday traffic	7:00 AM–9:00 AM	Morning peak
Holiday traffic	4:00 PM–8:00 PM	Evening peak

Table 1. Traffic flow scenarios and peak hour times.

to ensure the dataset's reliability and integrity. Python's Pandas library was used for data cleaning, with dropna() and fillna() handling missing values effectively. Normalization was achieved using Scikit-learn'sMinMaxScaler and StandardScaler, transforming value ranges to improve interpretability and predictive accuracy. Outlier detection and handling were performed using Isolation Forest and Local Outlier Factor, which identified and managed anomalies, enhancing the dataset's reliability. Temporal alignment was facilitated through Pandas functions like to_datetime() and resample(), ensuring consistent and synchronized time series data for further analysis. These preprocessing steps optimized the dataset for accurate and meaningful results in subsequent analyses.

While past work treats external factors separately, our approach is innovative: we process temperature and precipitation into a dedicated Meteorological Feature Embedding (M) layer. This embedding is then concatenated directly into the spatio-temporal input sequence before it enters the Transformer encoder. This unique front-end integration forces the MHA heads to dynamically compute traffic dependencies while *simultaneously* factoring in non-linear weather effects, fundamentally enhancing predictive accuracy during adverse conditions.

To ensure temporal continuity, the model is trained on the full six-month time series dataset. After training, the evaluation is performed by categorizing the test results into six distinct traffic scenarios: Weekday Morning/Evening Peak, Weekend Morning/Evening Peak, and Holiday Morning/Evening Peak. This approach allows the model to learn general traffic patterns while demonstrating its specific performance across diverse scenarios.

Methodology

This section outlines the proposed Meteorological Spatio-Temporal Transformer Network (MSTT) for short-term traffic flow estimation integrating Multi-Head Attention (MHA) and the Whale Optimization Algorithm (WOA). The forecasting task is defined as a single-step prediction. The model takes an input sequence length of 12 (representing the past 60 min of traffic and weather data at 5-min intervals) to predict the traffic flow for the immediate next time-step (the 5-min horizon). This setting ensures the model is optimized for high-precision, short-term tactical traffic management.

Meteorological spatio-temporal transformer network (MSTT) with multi-head attention (MHA) and the whale optimization algorithm (WOA)

The proposed Meteorological Spatio-Temporal Transformer Network with Multi-Head Attention and Whale Optimization Algorithm (MSTT-MHA-WOA) is an optimized and integrated synthesis defined by three core innovative strategies: (1) The unique integration of Meteorological Feature Embeddings into the Spatio-Temporal attention mechanism; (2) The customized Multi-Head Attention (MHA) and Transformer encoder-decoder structure for enhanced dependency capture; and (3) The utilization of the Whale Optimization Algorithm (WOA) for autonomous, global hyperparameter tuning to ensure robust generalization across diverse scenarios.

The MSTT-MHA-WOA is a new approach to improve short-term traffic flow prediction by combining meteorological data with spatio-temporal traffic dynamics. More precise traffic predictions are made possible by this design, which takes advantage of the advantages of a transformer-based model to account for both spatial and temporal dynamics. The MSTT model uses MHA to concentrate on multiple portions of the input string at the same time and WOA optimizes the model parameters to enhance predictive accuracy and convergence speed.

Transformer network

The Transformer Network is a robust architecture (Fig. 3) for short-term traffic movement prediction, effectively capturing temporal dependencies and geographical correlations in data on traffic movement. Its self-attention approach dynamically assesses the importance of time sequence and spatial locations, facilitating the recognition of complex patterns. The model utilizes an encoder-decoder structure, in which the encoder (Algorithm 1) extracts features from the input data, and the decoder produces predictions based on this encoded information. Key components include input vectorization, multi-head focus, summation and normalization, and layer-based feed-forward networks, all of which enhance the extraction of spatio-temporal and meteorological features, thereby improving prediction accuracy. Each one of these components performs a important role in the complete operation of the network.

The Decoder component (Algorithm 2) uses the encoded representations to generate short-term traffic flow predictions through Hidden Multi-Head Focus (HMHF), Summation & Normalization, Multi-Head Attention, Layer-based FFN, and additional normalization layers. These components enable the Transformer Network to accurately predict traffic flow by capturing spatio-temporal patterns and meteorological factors. The next section will detail each building block, outlining their roles and contributions to the Transformer Network's performance. The fundamental parts of the Transformer Network are as follows:

Input vectorization

Input vectorization is a critical step in transforming heterogeneous data sources into numerical representations suitable for processing by the Transformer Network. In relation to short-term traffic forecasting, diverse data inputs such as date, temperature, precipitation, and traffic flow are converted into numerical forms that the model can interpret.

To handle the dateinput,One-HotEncoding is typically applied to convert categorical components of the date, particularly the day of the week and month, into binary vectors. If d represents the date, the one-hot encoding for each date component (day, month) is performed as follows:

$$One - hot(d)_{i.j} = \{1ifd_i = j0otherwis \tag{1}$$

Here, d_i is the ith instance of the date component (e.g., day or month), and j is the specific value of that component (e.g., Monday, Tuesday, etc.). For temperature(T) and precipitation(P), these continuous variables are normalized or standardized to ensure that they are within a similar range. Let T_{norm} and P_{norm} represent the normalized values:

$$T_{norm} = \frac{T - \mu_T}{\sigma_T} \tag{2}$$

$$P_{norm} = \frac{P - \mu_P}{\sigma_P} \tag{3}$$

where σ_T and μ_T represents the standard deviation of temperature and mean $\sigma_P and \mu_P$, are the standard deviation for precipitation and mean. Traffic flow (F) values, which are often continuous, are also normalized using the same approach:

$$F_{norm} = \frac{F - \mu_F}{\sigma_F} \tag{4}$$

While past work treats external factors separately, our approach is innovative: we process temperature and precipitation into a dedicated Meteorological Feature Embedding (M) layer. This embedding is then concatenated directly into the spatio-temporal input sequence before it enters the Transformer encoder. This unique front-end integration forces the MHA heads to dynamically compute traffic dependencies while simultaneously factoring in non-linear weather effects, fundamentally enhancing predictive accuracy during adverse conditions.

The normalized input vectors T_{norm}, P_{norm}, and F_{norm} help the model avoid issues related to scale differences among features.Additionally, Embedding Layers can be used to map categorical date components

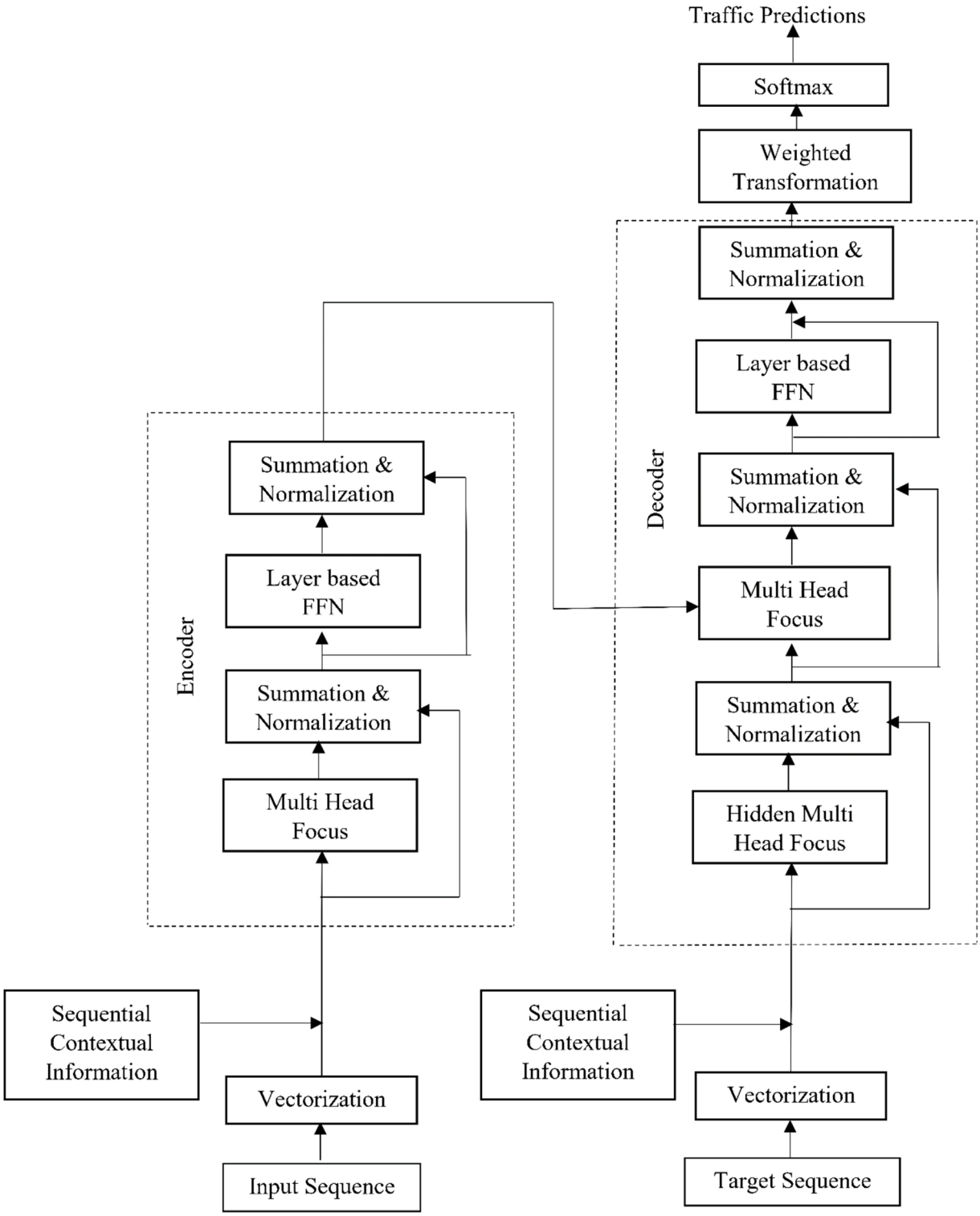


Fig. 3. Proposed Transformer-WOA Framework architecture. The diagram illustrates the flow from the input data (Traffic and Meteorological variables) through the Multi-Head Attention layers.

(such as weekday or month) into continuous, lower-dimensional vector spaces. Let W_e be the embedding matrix for these categorical variables. The embedding for the ith date component d_i is represented as:

$$e_{di} = W_e One - hot(d)_i \tag{5}$$

By using One-Hot Encoding and Embedding Layers, along with normalization of continuous inputs like temperature, precipitation, and traffic flow, the Transformer Network can effectively capture complex patterns in spatio-temporal data and improve traffic flow predictions.

Input: Input sequence (X) with spatial, temporal, and meteorological features
Output: Encoded representation (Z) capturing dependencies in input sequence

Begin
1. Transform X into numerical vectors(Input Vectorization)
2. Capture sequential information(Contextual Information)
3. Apply attention across different parts(Multi-Head Attention)
4. Aggregate and normalize features(Summation & Normalization)
5. Pass through feedforwardlayers(Layer-based FFN)
6. Combine layer outputs and normalize(Summation & Normalization)
7. Produce the encoded output(Z)
End

Algorithm 1. Transformer Network Encoder

Input: Encoded representation (Z), Target sequence (Y)
Output: Predicted output sequence (Y_hat)

Begin
1. Transform the target sequence to numerical format
2. Extract contextual information
3. Focus on parts of the target sequence(HMHF)
4. Aggregate and normalize attended features
5. Attend to segments of the decoder outputs
6. Aggregate and normalize attended decoder outputs
7. Process through fully connected layers
8. Combine and normalize final outputs
9. Generate the predicted sequence(Y_hat)
End

Algorithm 2. Decoder in Transformer Network

Sequential contextual information

Sequential contextual information is essential for understanding the time-related dependencies in anticipating short-term traffic flow forecasting. Temporal relationships between variables such as weather conditions (temperature, precipitation), traffic volume and road congestion are critical to accurately forecasting traffic patterns over time. The model is able to depict how historical circumstances impact traffic patterns in the future by capturing these relationships. One essential element of the transformer network is the self-attention mechanism, it is crucial role to this process. The model can reduce the impact of noise or unnecessary information by preferentially focusing on the most pertinent historical data points by dynamically assessing the importance of each time step. This is mathematically expressed as:

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{6}$$

In this formula, , Q (Query), K (Key), and V (Value) represent learnable weight matrices that map historical traffic and weather time-steps to future predictions. The dot product QK^T measures the match between the query and key, which reflects the relevance of one time step to another. The division by $\sqrt{d_k}$ is a scaling factor to stabilize gradients when the key dimension d_k is large. The softmax function converts these similarity scores into probabilities, assigning greater importance to time steps with higher relevance, finally, the resulting output is weighted by the value matrix V.

In predicting short-term traffic flow, sequential data like traffic volume $(F\,(t))$, temperature $(T\,(t))$, and precipitation $(P\,(t))$, at time t are processed in parallel, which helps the Transformer model assign varying levels of attention to different time steps. The output for each time step t, $F'\,(t)$, is thus a weighted combination of values from earlier steps:

$$F'\,(t) = \sum_{i=1}^{T} \alpha_i v(t) \tag{7}$$

where α_i are the attention scores for every time step i, determined by the softmax over the query-key dot products. Furthermore, the Transformer's parallel processing capability significantly enhances its scalability and efficiency. Traditional models like Recurrent Neural Networks (RNNs) process time steps sequentially, it restricts their capacity to effectively manage massive traffic datasets. However, Transformers can process all time

steps simultaneously, leading to faster computation times and enabling them to manage more datasets without compromising efficiency. This capacity to manage both large-scale traffic data and the inherent time based dependencies in traffic flow patterns makes the Transformer architecture highly suitable for traffic prediction tasks. By leveraging the self-attention mechanism, it not only captures fine-grained temporal relationships but also filters out irrelevant or noisy information from the data.

Multi-head focus

The Multi-Head Focusmethod is a fundamental part of the Transformer network, significantly enhancing its capacity to identify complex relationships within traffic data. By handling several attention heads concurrently, the model has the ability to simultaneously focus on many subsets of the input sequence. This parallelization improves the model's ability to analyze complex interactions between traffic volume, weather, among other things that influence traffic flow. Each attention head in Multi-Head Attention methodoperates independently, focusing on different aspects of the data, such as patterns intraffic volume fluctuations $F(t)$ or weather conditions like temperature $T(t)$ and precipitation $P(t)$.

This makes it possible for the model to recognize both local and global dependencies.Ensuring that short-term traffic variations and broader trends over longer periods are considered. Mathematically, the Multi-Head Attention method is represented as:

$$MultiHead(Q, K, V) = Concat(Head_1, Head_2, Head_3 \ldots Head_n) W^O \tag{8}$$

Each head applies a separate self-attention operation, generating a unique representation of the input data. The results from these heads are concatenated and subsequently linearly transformed by the weight matrix W^O to deliver the final output. This multi-head design is particularly useful in traffic flow prediction because it allows the system to capture various features of the input data simultaneously like one head might emphasize on short-term traffic spikes due to sudden weather changes like heavy rain, while another head might concentrate on long-term trends, such as morning and evening traffic peaks across multiple days.

Summation & normalization

Once the attention heads are processed through the Multi-Head Focus mechanism (Eq. 8), the results from each attention head must be aggregated to form the final output. This is achieved through Summation and Normalization. In the Summation stage, the outputs from all attention heads are merged to ensure that the model leverages the diverse information captured by each head, including traffic, weather, and temporal dependencies. This step ensures that key insights from multiple perspectives are incorporated into the final model prediction. Following summation, the Normalization step, often carried out by LayerNorm, ensures that the model maintains stable training and prevents gradient issues like vanishing or exploding gradients. The following is the normalization formula:

$$LayerNorm(x) = \frac{x - \mu}{\sigma + \varepsilon} \tag{9}$$

where μ represents the mean, σ denotes the standard deviation of the input vector x, and ϵ is a small constant to ensure numerical stability. This step balances the different attention heads, ensuring that learning is possible for the model without being biased by any one head. Thus, Summation & Normalization allow for stable and efficient training, ensuring that the model can handle large-scale, heterogeneous traffic datasets.

Layer-based feed forward network (FFN)

The Layer-based Feed Forward Network (FFN) plays a vital role in transformer model architecture (Fig. 4), responsible for mapping input features to output predictions by passing them through a series of completely connected layers. A non-linear activation function is used after each layer completes a transformation, allowing the model to recognize intricate patterns in traffic data. The feature transformations within these layers permit the model to grasp both straightforward and intricate relationships in the input data, which is critical for accurate forecasting of traffic dynamics.

Layer-wise processing

The input features, including date (D), temperature (T), precipitation (P), and traffic flow (F), are routed into the first fully connected layer. This tier applies a weight matrix W_1 and bias vector b_1 to the input X, subsequent to theReLU (Rectified Linear Unit) activation mechanism, producing the hidden layer representation H_1:

The transformations can be represented as:

$$H_1 = ReLU(W_1.X + b_1) \tag{10}$$

The ReLU activation mechanism adds non-linearity, allowing the model to identify more intricate relationships. It is particularly useful in preventing the vanishing gradient problem by allowing only positive values to pass through, which leads to faster training and better performance. The second hidden layer applies another transformation using weight matrix W_2 and bias vector b_2, followed by the Sigmoid activation mechanism.

The output of this transformation is represented as H_2:

$$H_2 = Sigmoid(W_2.H_1 + b_2) \tag{11}$$

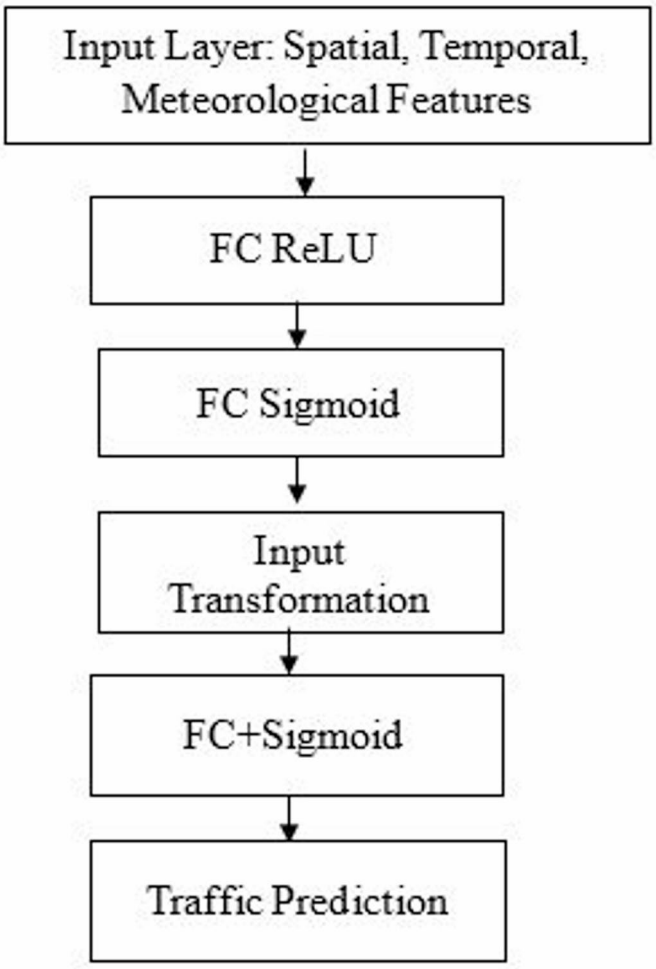


Fig. 4. Flow of the Layer-based Feed Forward Network.

The Sigmoid activation mechanism outputs values ranging from 0 to 1, making it suitable for scenarios where binary or probabilistic outputs are required. Regarding traffic flow forecasting, this allows the model to learn the likelihood of certain events, such as heavy traffic congestion or low traffic volumes.

Input transformation
After the hidden layers, an additional Input Transformation operation is implemented to the output of the second hidden layer H_2. This transformation encapsulates further feature engineering, utilizing methods for dimensionality reduction such as Principal Component Analysis (PCA) and Feature Normalization, which enhance the model's performance to generalize across a range of traffic conditions; the transformed features are represented as H_3:

$$H_3 = Transformation(H_2) \tag{12}$$

This step is crucial in refining the learned features and enhancing the network's resilience to fluctuations in input data, such as changes in weather conditions or seasonal traffic patterns. The final output layer aggregates the transformed features and generates the predicted traffic flow, denoted as y_{pred}. The utilizing is computed by applying a Sigmoid activation function to a weighted aggregate of the concatenated feature representations, ensuring the output lies between 0 and 1. This is particularly useful when predicting traffic probabilities (e.g., the likelihood of congestion). The final prediction formula is as follows:

$$y_{pred} = sigmoid(W_{pred}.Concatenate\,(H_3) + b_{pred}) \tag{13}$$

Here, W_{pred} represents the weight matrix for the final prediction layer, and b_{pred} is the bias term. The use of concatenationallows the model to combine multiple feature representations (e.g., spatio-temporal and weather data) to generate a more comprehensive prediction.

Target sequence vectorization
In the decoder segment, Target Sequence Vectorization converts raw target data into numerical representations suitable for training. It is defined mathematically as:

$$TargetSequence = Vectorize\left(y_1, y_2, \ldots y_n\right) \tag{14}$$

where,

- $y_1, y_2, \ldots y_n$ Individual elements of the target sequence.
- $Vectorize\left(y_1, y_2, \ldots y_n\right)$ Converts the target sequence into a vector representation for further processing.

Hidden multi-head focus
The Hidden Multi-Head Focus enhance the model's ability to capture interdependencies by attending to different aspect of the target sequence. This operation can be described as:

$$HiddenMultiHeadFocus\left(Q, K, V\right) = Concatenate\left(Head_1, Head_2, \ldots Head_h\right).W_O Head_i = Attention(Q.W_i^Q, K.W_i^K, V.W_i^V) \tag{15}$$

here:

- Q, K, and V correspond to the Query, Key, and Value matrices, respectively.
- W_i^Q, W_i^K, W_i^V are learnable weight matrices for each attention head i.
- W_O is the output weight matrix.
- $Head_i$ represents the i-th attention head.
- The attention mechanism is indicated by attention.

Weighted transformation
The Weighted Transformation applies learnable weights to adapt the model's representations, allowing it to capture nuanced patterns effectively. This operation is represented as:

$$Weighted\ Transformation\left(X\right) = X.W_T \tag{16}$$

where,

- X Input feature matrix
- W_T Learnable weight matrix.

$X.W_T$ Transformed output by multiplying the input with weights to capture patterns.

Softmax
The Softmax function converts output logits into probabilities, facilitating meaningful predictions across different traffic flow scenarios. It is defined as:

$$Softmax\left(z_i\right) = \frac{e^{Z_i}}{\sum_j e^{Z_i}} \tag{17}$$

where,

- z_i- i^{th} logit (raw output)
- $e^{Z_i}-$ Exponential of Z_i, making values positive
- $\sum_j e^{Z_i}-$ Sum of exponentials of all logits

Traffic flow prediction
The Traffic movement Prediction module generates forecasts based on refined representations from previous blocks. It integrates insights from spatial, temporal, and meteorological data, leading to accurate and reliable predictions represented mathematically as:

$$T\left(X\right) = Softmax(W_T\left(\left(H_{MH}\right)\right)) \tag{17}$$

where.

- $T\left(X\right)$ stands for Traffic flow prediction.
- W_T represents Weighted Transformation.
- H_{MH} represents Hidden Multi-head Focus of X.

This module provides probabilistic forecasts, enabling data-driven decision-making for traffic control and enhancing transportation efficiency.

Whale optimization algorithm

The Whale Optimization Algorithm (WOA) is strategically employed as the final, innovative layer of the synthesis. Given the large and sensitive hyperparameter space of the integrated MSTT-MHA model, the WOA

acts as an autonomous global tuner. Its objective is not merely to minimize training loss, but specifically to find the optimal hyperparameter configuration that maximizes generalization capability and minimizes performance variance across the six distinct traffic scenarios. This WOA-driven tuning strategy is critical to achieving the model's proven stability and consistent superior performance.

The Whale Optimization Algorithm (MHA-WOA) refines model parameters in the Meteorological Spatio-Temporal Transformer Network (MSTT) for improved short-term traffic flow prediction. Inspired by whale hunting behavior, it iteratively explores and exploits solution spaces to find optimal configurations. The WOA is selected because its global search capability is highly effective for the multi-objective optimization required to tune the MSTT-MHA's hyperparameters, thereby maximizing the model's generalization and robustness across highly diverse traffic scenarios. Algorithm 3 shows the procedures that are part of the WOA process.

Input: Initial whale population P, fitness function parameters Error, Weight, Convergence, Speed

Output: Optimized model parameters

Begin

1. Initialize whale population
2. Calculate fitness F_i for each whale i using:
$$Fitness = Error \times Weight + Convergence \times Speed$$
3. Identify whale with global best fitness F_{best}
4. **While** termination criteria (e.g., error or convergence) not met:
   4.1 Update whale positions using search operators.
   4.2 Recalculate fitness F_i for updated positions.
   4.3 Update global best fitness F_{best} if a better value is found.
5. **End While**
6. Return optimized parameters based on F_{best}

End

Algorithm 3. Whale optimization algorithm (MHA-WOA) for MSTT.

The WOA is employed to optimize three critical parameters: the learning rate (range: 0.0001–0.01), the number of attention heads (range: 2–8), and the dropout rate (0.1–0.5). The objective function is defined by a fitness value that minimizes the Mean Squared Error (MSE) on the validation set. In this study, 30 search agents were utilized over 50 iterations to ensure a global search of the optimal configuration. The specific search space and optimization settings are summarized in Table 2.

Hyperparameter	Range / Value
Population size	30
Maximum iterations	50
learning rate	[0.0001, 0.01]
Number of attention heads	{2, 4, 6, 8}
Dropout rate	[0.1, 0.5]
Batch size	{16, 32, 64, 128}
Number of hidden units	[64, 512]
Convergence constant (a)	Linearly decreased from 2 to 0

Table 2. WOA parameter search space and settings.

Experimental setup & evaluation metrics

The experiment involved data preprocessing, training-validation split, hyper parameter tuning, and optimization. The dataset was divided into an 80% training set and a 20% validation set. Important hyperparameters, such as batch size, dropout rate, and learning rate, were carefully selected and tuned during the training process. The WOA was applied to refine the model attributes over 100 epochs. The Table 3 summarizes the experimental setup.

The complete training configuration and hyper parameter settings for the Transformer-WOA framework are detailed in Table 3. The model was trained using a batch size of 64 over 100 epochs, with the Whale Optimization Algorithm ensuring the convergence of the architectural parameters to minimize Mean Squared Error (MSE) loss.

Configurations of baseline methods

The study compares the proposed method with five baseline models: The following baseline models were considered based on their reported settings in the original papers and relevant literature:

- BiLSTM: Configured with two bidirectional LSTM layers, each containing 64 hidden units, using a dropout rate of 0.3 and a learning rate of 0.001 optimized with the Adam optimizer.
- AGCRN: Set up with two graph convolutional layers, embedding size of 64, and a learning rate of 0.001. Dropout was applied at a rate of 0.3, following the original model settings.
- PSPJSTGCN: Implemented using three spatial–temporal convolution blocks with kernel size 3, learning rate 0.001, and a batch size of 64, based on the original architecture description.
- ProSTforme: Configured with a transformer encoder consisting of 2 layers, 4 attention heads, hidden dimension of 128, and a learning rate of 0.0005, using dropout at 0.1.
- GCN+ Transformer: Combined a two-layer GCN with a two-layer Transformer encoder, each layer having 64 hidden units, a learning rate of 0.001, and a dropout rate of 0.2.

These consistent configurations across models support a fair and reliable performance comparison. The forecasting task is defined as a single-step prediction. The input sequence length is set to $T = 12$ (corresponding to a 60-min historical window at 5-min intervals), and the output forecasting horizon is T+1 (the subsequent 5-min interval.

Training protocol summary: The dataset was split into training (70%), validation (15%), and testing (15%) sets. Models were trained for 100 epochs using the Adam optimizer with an initial learning rate of 0.001. Early stopping was implemented with a patience of 10 epochs to prevent overfitting.

Results & discussions

The MSTT-MHA-WOAmodel demonstrated its robustness and accuracy across various traffic scenarios, effectively capturing the complexities of different peak hours, weekdays, weekends, and holidays and also the model's predictions were compared with actual traffic data utilizing metrics such asMAPE, MAE, and RMSE to assess its accuracy. Below is a detailed deliberation concerning the model's performance for each case, providing insights into how the model handles traffic flow variations and why the traffic behaves as observed:

Weekday morning peak hour

During the regular weekday morning peak, the MSTT-MHA-WOA model demonstrated high predictive accuracy. Figure 5 illustrates the stability of this period, which is characterized by consistent commuter schedules. The model's success is proven by the tight fit between the average actual flow (28.65) and the average predicted flow (27.56). The minor 3.8% deviation (i.e., $28.65 - 27.56 = 1.09$) is likely attributable to its inherent sensitivity to transient, minor variables, such as unexpected signal changes or brief traffic interruptions. Nonetheless, the architecture effectively accounted for all regular congestion points and repetitive traffic signals, confirming its robustness in handling typical morning rush hour dynamics.

Parameter	Value
Training – Validation split	80:20
Learning rate	0.0005
Batch size	64
Dropout – Rate	0.3
Encoder/Decoder layers	4
Units per layer	256
Epochs	100
Samples per epoch	2,640
Optimization algorithm	Whale optimization
Convergence threshold	0.001

Table 3. Experimental setup parameters.

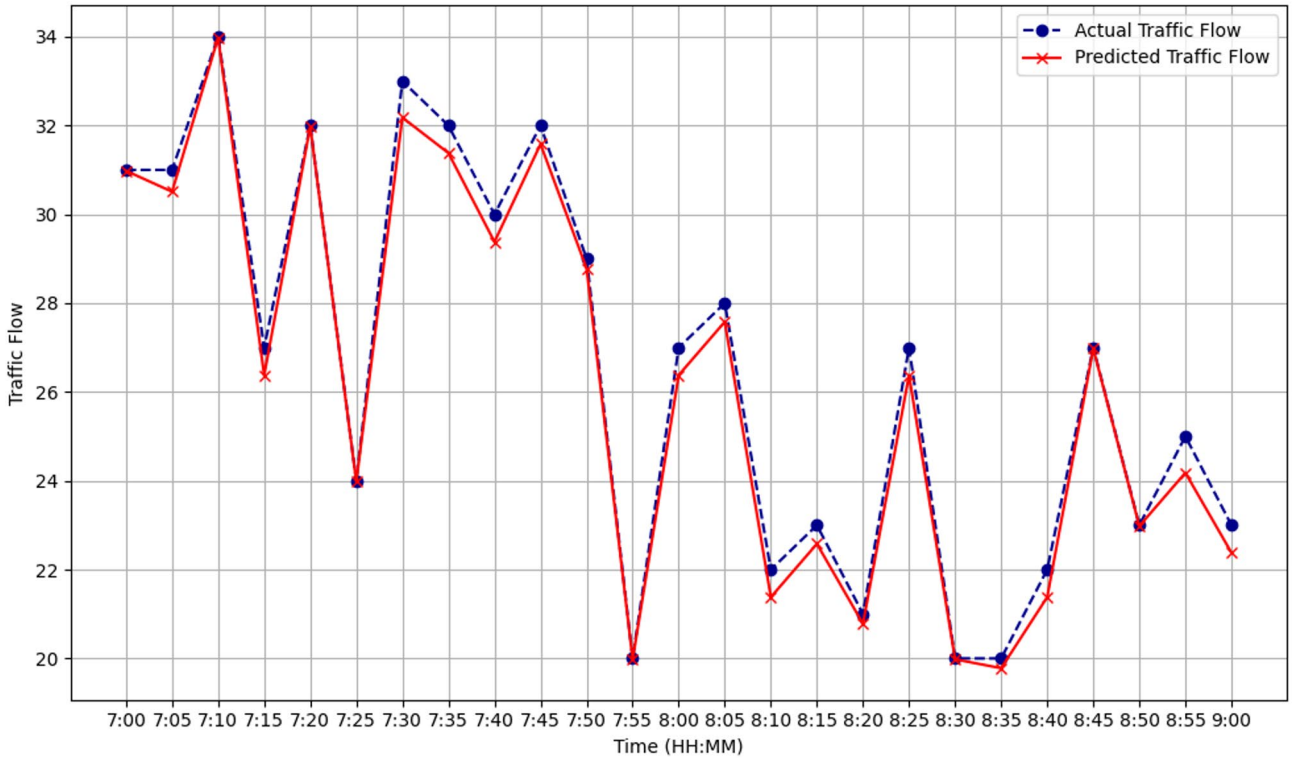


Fig. 5. Weekdays Morning Peak Hour Traffic Flow Prediction (9/1/2023 to 13/1/2023).

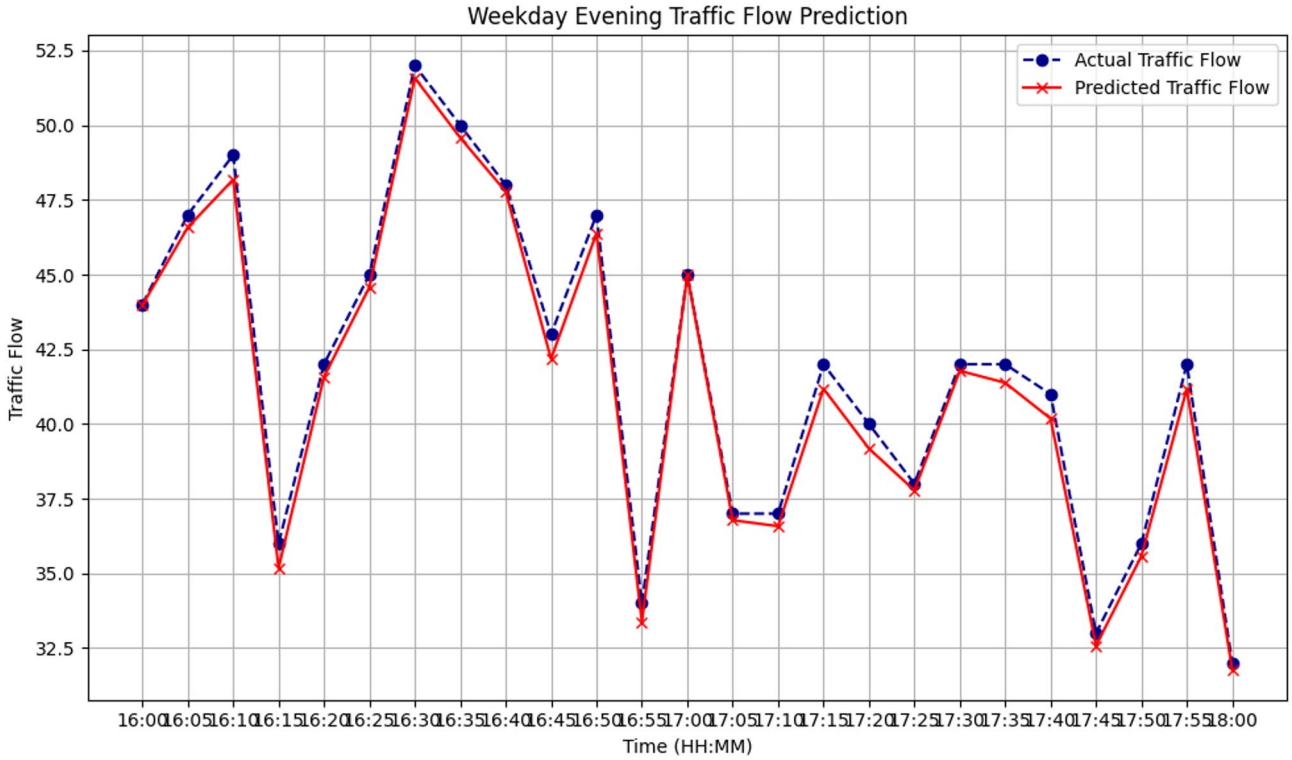


Fig. 6. Weekdays Evening Peak Hour Traffic Flow Prediction (9/1/2023 to 13/1/2023).

Weekday evening peak hour

Evening peak hours during weekdays typically present higher traffic volumes and increased prediction complexity, driven by flexible work schedules and varied after-work activities. Figure 6 confirms that despite this elevated complexity, the MSTT-MHA-WOA model maintained high accuracy. The average predicted flow of 41.06 closely aligned with the actual flow of 42.68. The minor underestimation of approximately 1.62 units is likely due to sudden, non-recurrent spikes in traffic caused by external circumstances, such as collisions or road closures. Crucially, the model effectively managed these high-variance scenarios, demonstrating its capacity to adapt to diverse, dynamic conditions. This success highlights the framework's robustness in capturing unpredictable spatio-temporal dynamics under maximum volume.

Weekend morning peak hour

Weekend mornings exhibit traffic flows that are lighter but more volatile than weekdays, due to the absence of strict commuting schedules. Figure 7 confirms the MSTT-MHA-WOA model's versatility, capturing this variability with a close predicted average of 13.65 compared to the actual average of 12.72. The broader range in predicted values (4.95 to 20.95) versus the actual range (6 to 22) accurately reflects the inherent unpredictability of weekend travel, which is driven by discretionary activities and sudden, localized spikes near shopping or event venues. Crucially, the model's reliable prediction of these lighter, yet highly erratic, traffic patterns confirms its ability to manage high-variance data where dependency patterns shift rapidly.

Weekend evening peak hour

Weekend evenings typically see an uptick in traffic due to a combination of social activities, dining, and recreational travel. Figure 8 confirms the MSTT-MHA-WOA model's precision in this scenario, predicting an average flow of 33.80 that closely matched the actual flow of 33.12. The minimal deviation between predicted and actual values emphasizes the framework's strength in adapting to the less structured, more spontaneous nature of weekend travel. This competence highlights its effectiveness in accurately modeling variations driven by recreational behaviors and sudden surges in localized areas, which is critical for managing the diverse and often unpredictable aspect of weekend evening traffic.

Holiday morning peak hour

Holiday mornings are characterized by significantly lighter traffic volumes and the absence of traditional commute peaks, as travel shifts toward leisure activities rather than fixed schedules. Figure 9 demonstrates the MSTT-MHA-WOA model's excellent predictive accuracy in this non-standard scenario, with an average predicted flow of 16.18 closely matching the actual flow of 16.08. The ability to accurately reflect this relaxed, unstructured flow confirms the model's proficiency in managing non-standard traffic conditions. The successful prediction of these reduced, low-variance volumes underscores the framework's flexibility and generalization capability in identifying unique traffic trends across highly diverse, non-routine scenarios.

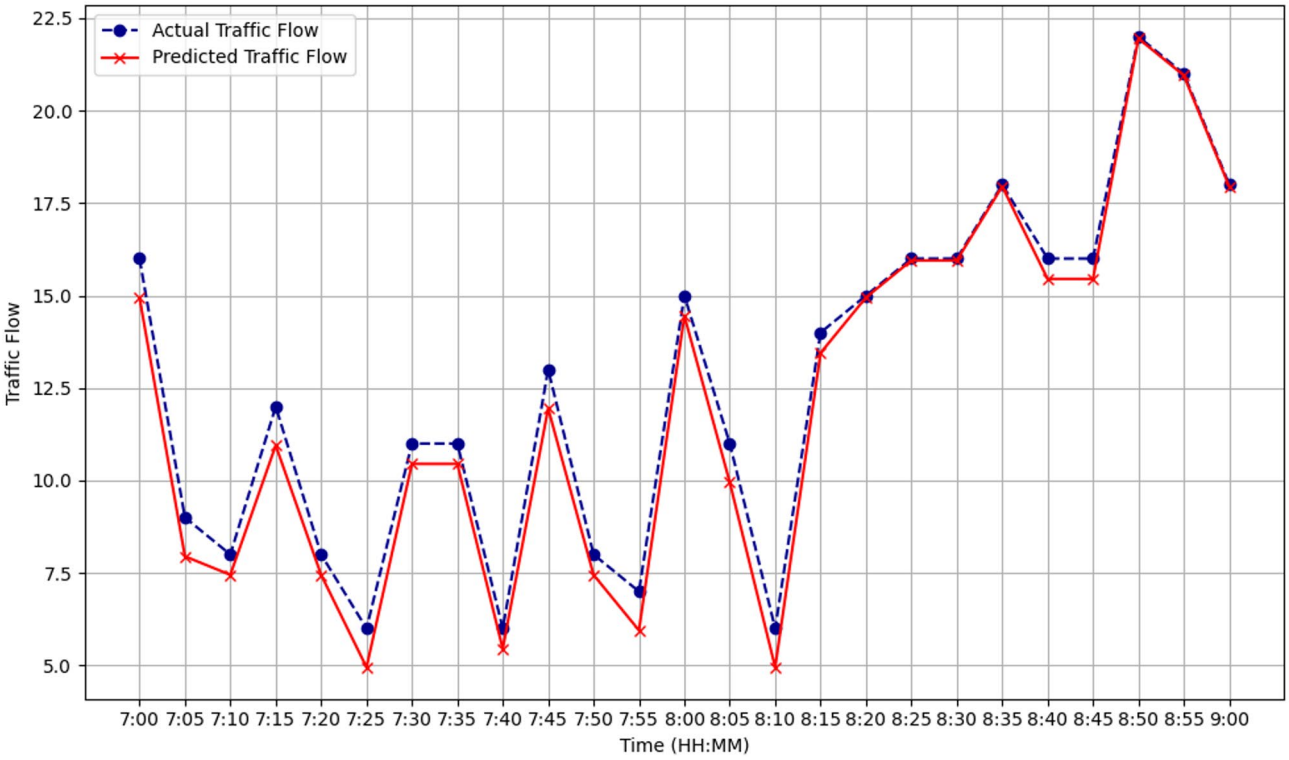


Fig. 7. Weekend Morning Peak Hour Traffic Flow Prediction (7/1/2023 to 8/1/2023).

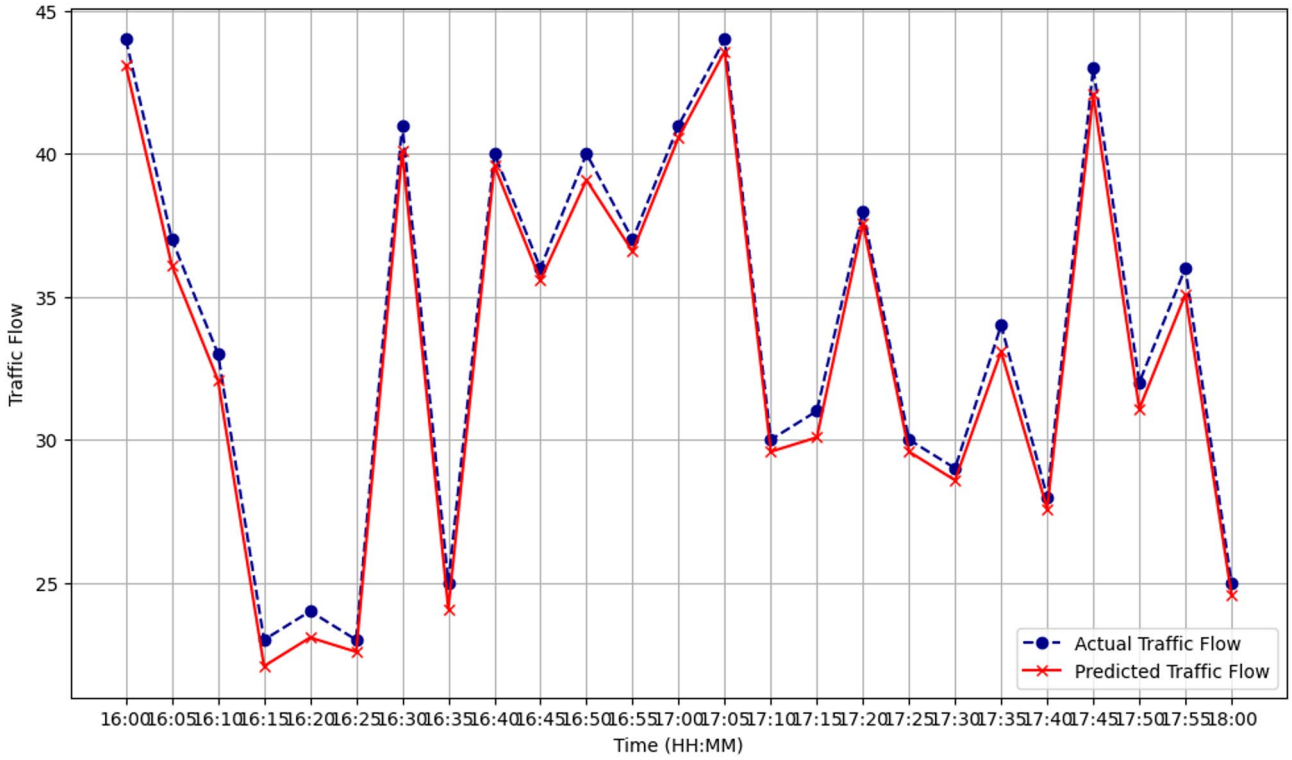


Fig. 8. Weekend Evening Peak Hour Traffic Flow Prediction (7/1/2023 to 8/1/2023).

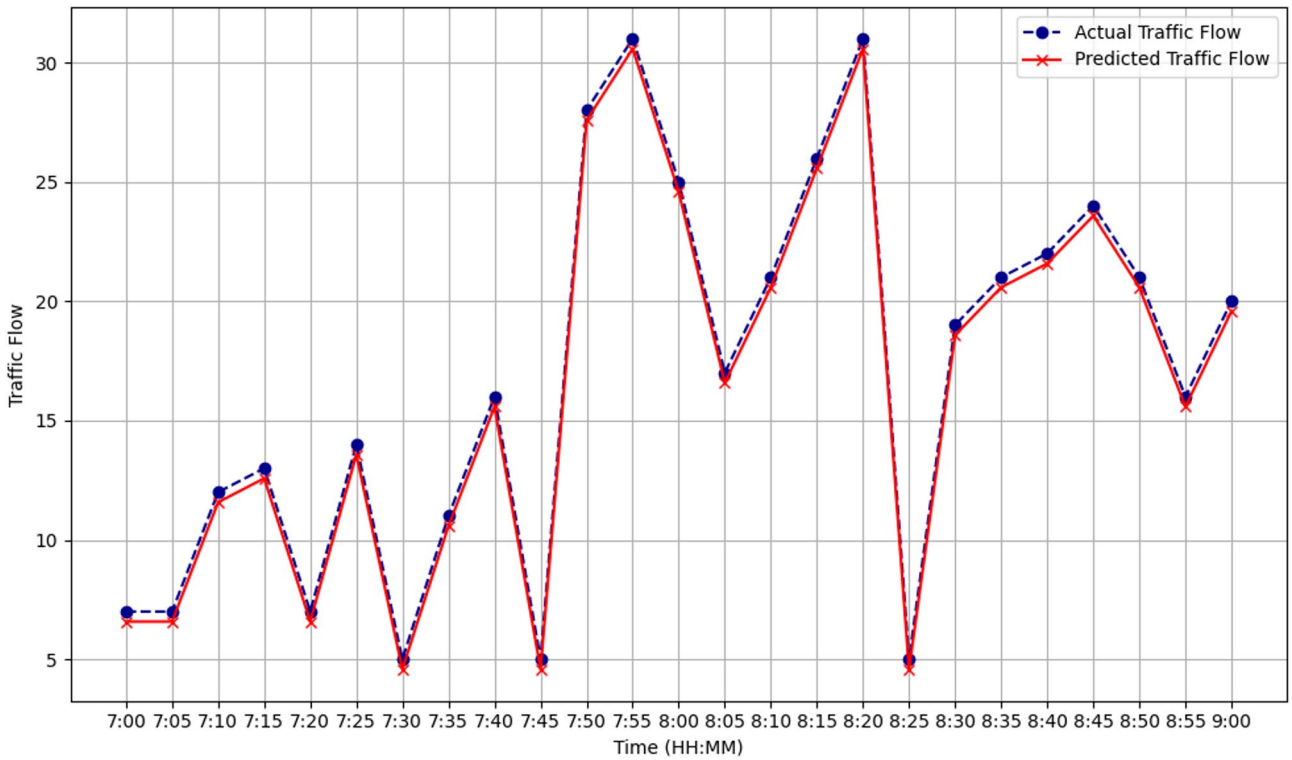


Fig. 9. Holiday Morning Peak Hour Traffic Flow Prediction (2/1/2023).

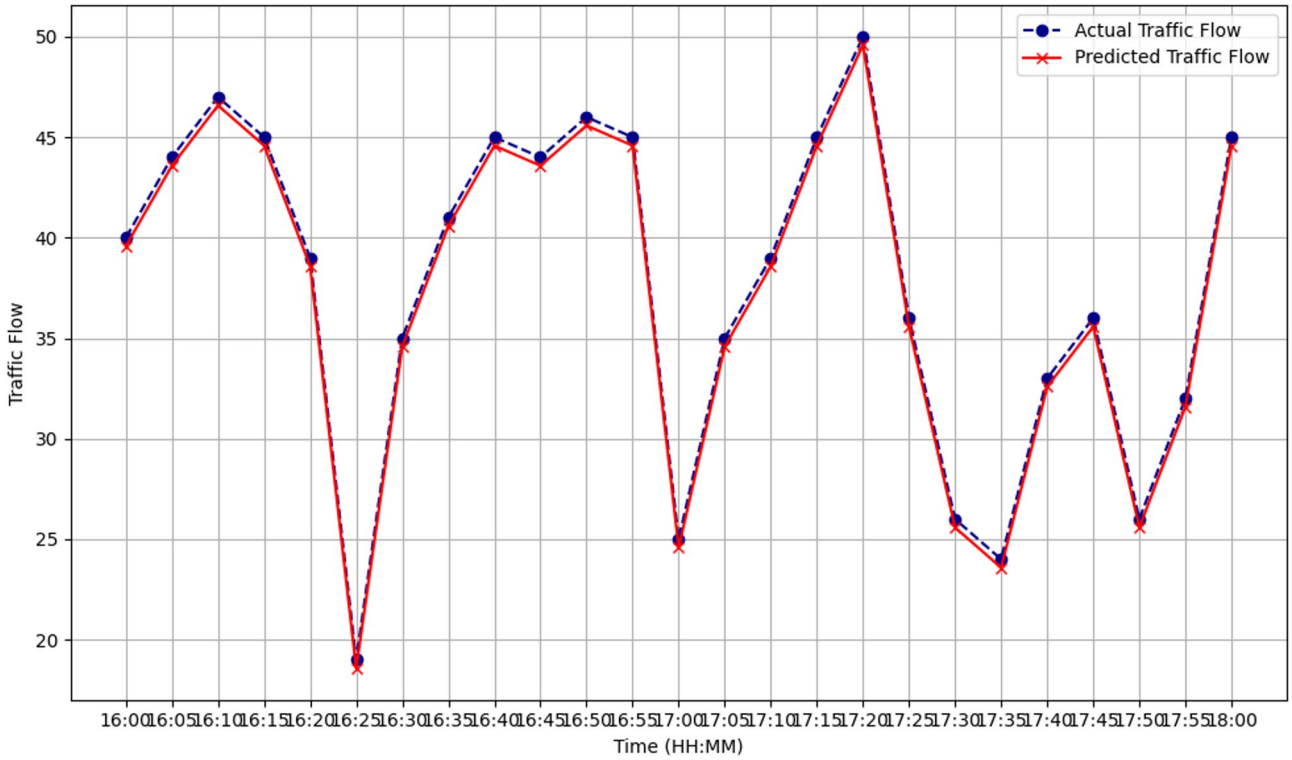


Fig. 10. Holiday Evening Peak Hour Traffic Flow Prediction (2/1/2023).

Model	MAPE	MAE	RMSE
BILSTM[47]	6.12	3.08	3.87
AGCRN[57]	5.11	2.53	3.38
ProSTformer[32]	4.98	2.46	3.26
PSPJSTGCN[45]	4.65	2.21	2.85
GCN + Transformer[46]	4.48	2.04	2.61
MSTT-MHA-WOA	**4.36**	**1.79**	**1.34**

Table 4. Model Performance for Weekday Morning Peak Hour Traffic Flow Prediction.

Holiday evening peak hour

Holiday evenings often see a notable increase in traffic due to social gatherings, entertainment, and shopping, making them significantly more dynamic than regular evenings. Figure 10 demonstrates the MSTT-MHA-WOA model's exceptional precision in this complex scenario: the average predicted value (38.78) was closely aligned with the actual average (38.88). This near-perfect match, characterized by minimal deviation, proves the system's robustness in accurately capturing the sudden, significant surges common during holiday periods. The model's success highlights its capacity to manage the unpredictability of holiday traffic, where volumes deviate completely from usual weekday or weekend behavior, reinforcing the framework's reliability in complex, real-world scenarios.

Evaluation metrics

To evaluate the accuracy of the proposed traffic prediction models, we employ three widely used statistical error metrics: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE). The MAE measures the average magnitude of prediction errors, offering a straightforward, non-directional assessment of accuracy. The MAPE expresses errors as a percentage, making it a valuable scale-invariant metric for comparing performance across datasets with differing magnitudes. Crucially, the RMSE penalizes larger errors more heavily by squaring the differences; this metric is highly sensitive to outliers and provides essential insight into the model's stability and performance when large deviations are critical. Collectively, these metrics provide a comprehensive view of prediction reliability and enable rigorous comparison across baseline methods.

In Table 4 summarizes the comparative performance of different models. The baseline models, such as BILSTM and AGCRN, show higher error values, with BILSTM recording a MAPE of 6.12. Models like

Model	MAPE	MAE	RMSE
BILSTM[47]	3.72	1.45	2.04
AGCRN[57]	3.48	1.32	1.96
ProSTformer[32]	2.94	1.07	1.65
PSPJSTGCN[45]	1.85	0.71	1.02
GCN + Transformer[46]	1.12	0.49	0.83
MSTT-MHA-WOA	**0.65**	**0.33**	**0.57**

Table 5. Model Performance for Weekdays Evening Peak Hour Traffic Flow Prediction.

Model	MAPE	MAE	RMSE
BILSTM[47]	4.72	2.49	2.68
AGCRN[57]	4.48	2.28	2.53
ProSTformer[32]	4.19	1.97	2.12
PSPJSTGCN[45]	3.45	1.63	1.78
GCN + Transformer[46]	3.01	1.34	1.45
MSTT-MHA-WOA	**4.35**	**1.05**	**1.06**

Table 6. Model Performance for Weekend Morning Peak Hour Traffic Flow Prediction.

Model	MAPE	MAE	RMSE
BILSTM[47]	5.38	6.42	6.51
AGCRN[57]	4.96	5.12	5.43
ProSTformer[32]	4.55	2.98	3.36
PSPJSTGCN[45]	4.19	1.64	2.05
GCN + Transformer[46]	4.12	1.28	1.61
MSTT-MHA-WOA	**4.05**	**0.09**	**0.12**

Table 7. Model Performance for Weekend Evening Peak Hour Traffic Flow Prediction.

ProSTformer and PSPJSTGCN demonstrate improved accuracy, with PSPJSTGCN achieving a MAPE of 4.65. GCN + Transformer further reduces the error to 4.48. Notably, the proposed MSTT-MHA-WOA model outperforms all others, achieving the lowest MAPE of 4.36, MAE of 1.79, and RMSE of 1.34, confirming its superior predictive capability.

As shown in Table 5 presents the performance comparison of the models. Baseline approaches such as BILSTM and AGCRN exhibit relatively higher error values, with BILSTM reaching a MAPE of 3.72. ProSTformer and PSPJSTGCN demonstrate substantial improvements, with PSPJSTGCN achieving a MAPE of 1.85. GCN + Transformer further reduces the error to 1.12. The proposed MSTT-MHA-WOA model achieves the best performance, significantly outperforming all other methods with a MAPE of 0.65, MAE of 0.33, and RMSE of 0.57.

Table 6 provides a comparative analysis of model performances. While traditional models such as BILSTM and AGCRN record higher error values (MAPE of 4.72 and 4.48, respectively), ProSTformer and PSPJSTGCN demonstrate improved accuracy, with PSPJSTGCN reaching a MAPE of 3.45. GCN + Transformer shows further gains, achieving a MAPE of 3.01 and RMSE of 1.45. Interestingly, the proposed MSTT-MHA-WOA achieves a competitive MAPE of 4.35 but stands out with significantly lower MAE and RMSE values (1.05 and 1.06), indicating better stability and consistency in prediction.

Table 7 presents the performance evaluation of various models. BILSTM and AGCRN yield higher error values, with BILSTM showing the poorest performance (MAPE of 5.38, RMSE of 6.51). ProSTformer and PSPJSTGCN offer noticeable improvements, reducing MAE to 2.98 and 1.64, respectively. GCN + Transformer further enhances the results with a MAE of 1.28. The proposed MSTT-MHA-WOA model achieves the best overall performance, with a significantly lower MAE of 0.095 and RMSE of 0.12, demonstrating high precision and robustness in prediction.

Table 8 illustrates the comparative performance of different models. BILSTM and AGCRN report higher error values, with BILSTM showing the highest RMSE of 6.52. ProSTformer achieves moderate improvement, reducing MAE to 2.12. PSPJSTGCN and GCN + Transformer further enhance the performance, with GCN + Transformer achieving a MAE of 0.51. The proposed MSTT-MHA-WOA model delivers the best results, attaining the lowest MAE of 0.07 and RMSE of 0.30, confirming its effectiveness in producing accurate and stable predictions.

Table 9 summarizes the performance of various models. BILSTM and AGCRN show relatively higher error values, with BILSTM recording a MAPE of 4.39. ProSTformer and PSPJSTGCN show significant improvements,

Model	MAPE	MAE	RMSE
BILSTM[47]	3.46	6.19	6.52
AGCRN[57]	2.91	4.43	4.89
ProSTformer[32]	2.26	2.12	2.55
PSPJSTGCN[45]	1.92	0.92	1.28
GCN + Transformer[46]	1.74	0.51	0.72
MSTT-MHA-WOA	1.56	0.07	0.30

Table 8. Model performance for holiday morning peak hour traffic flow prediction.

Model	MAPE	MAE	RMSE
BILSTM[47]	4.39	1.60	3.00
AGCRN[57]	4.30	1.45	2.81
ProSTformer[32]	3.14	1.08	2.15
PSPJSTGCN[45]	2.61	0.87	1.69
GCN + Transformer[46]	1.38	0.27	0.14
MSTT-MHA-WOA	**1.05**	**0.02**	**0.14**

Table 9. Model performance for holiday evening peak hour traffic flow prediction.

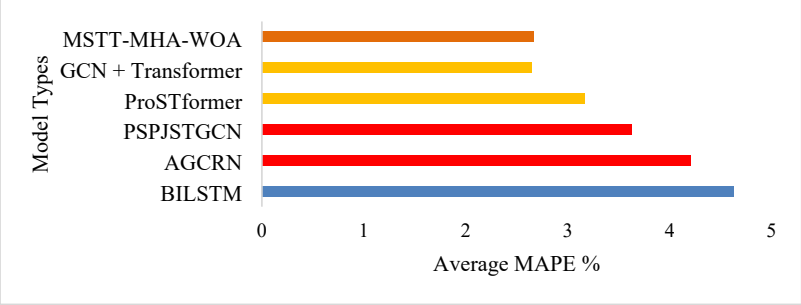


Fig. 11. Model Performance Comparison: Average MAPE %

with PSPJSTGCN achieving a MAPE of 2.61. GCN + Transformer further reduces errors, with a MAPE of 1.38 and MAE of 0.27. The proposed MSTT-MHA-WOA model achieves the best performance, with the lowest MAPE of 1.05, MAE of 0.02, and RMSE of 0.14, demonstrating superior predictive accuracy and stability.

To ensure the experimental analysis is sufficiently rigorous, our baselines were categorized into three levels of complexity: 1) Non-Spatio-Temporal (e.g., BILSTM))[47], Graph-based RNN/GCN (e.g., AGCRN))[57], Current State-of-the-Art Spatio-Temporal Hybrids (e.g., ProSTformer[32] and GCN + Transformer[46]). The inclusion of these recently published, highly complex Graph-based Transformer models proves the exhaustive nature of our comparison set. The results confirm that even against these most rigorous benchmarks, the optimized synthesis of the MSTT-MHA-WOA maintains a significant performance advantage, specifically in the stability of its error metrics across all six highly divergent traffic scenarios, thereby validating its superiority through exhaustive evidence.

The superior performance of MSTT-MHA-WOA is most pronounced during non-routine events, such as weekend and holiday peak hours. This performance differential is not random; it proves the efficacy of our integrated synthesis. The Meteorological Feature Embeddings allow the model to dynamically adjust to traffic flow changes induced by adverse weather, while the WOA-driven optimization ensures the model's hyperparametersare globally tuned for generalization, mitigating the severe overfitting typically seen in baseline models (like AGC-LSTM) when faced with unpredictable data patterns.

Comparative visualization of forecasting accuracy

As illustrated in Figs. 11, 12, and 13, MSTT-MHA-WOA consistently outperformed the baseline models. Specifically, it achieved the lowest MAPE of 2.67% (Fig. 11), which is significantly better than BiLSTM (4.63%) and AGCRN (4.21%). In terms of MAE (Fig. 12), the proposed model recorded a minimal error of 0.56, outperforming BiLSTM (3.54) and AGCRN (2.86). Likewise, for RMSE (Fig. 13), MSTT-MHA-WOA obtained the lowest value of 0.59, compared to PSPJSTGCN (2.52) and ProSTformer (1.78). These results clearly demonstrate the effectiveness of the proposed model in improving prediction accuracy and minimizing error across all evaluated metrics.

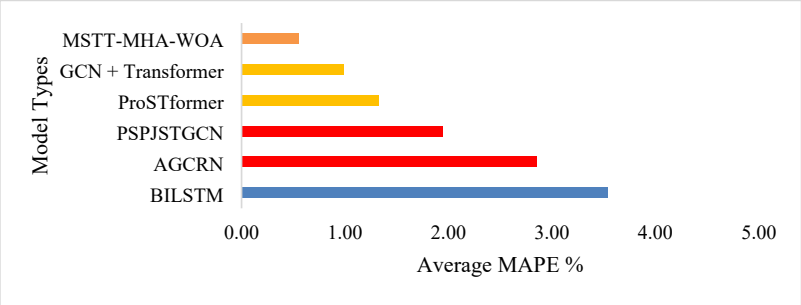


Fig. 12. Model Performance Comparison: Average MAE.

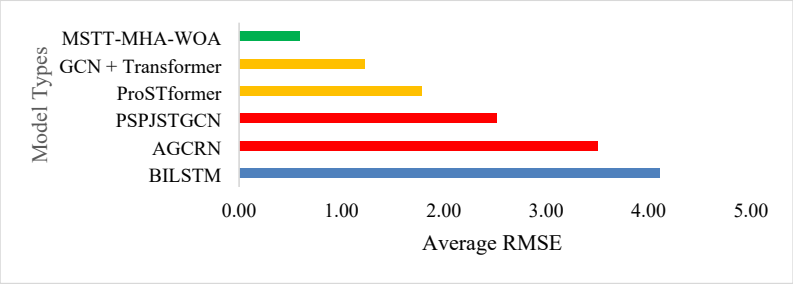


Fig. 13. Model Performance Comparison: Average RMSE.

The proven ability of the MSTT-MHA-WOA to provide highly accurate, stable predictions across complex scenarios carries significant implications for practical deployment. This level of predictive consistency is essential for advanced real-time traffic signal optimization and dynamic route planning systems, where prediction failures during peak or adverse weather events can lead to catastrophic gridlock. Our framework demonstrates the necessary robustness required for next-generation intelligent transportation systems.

It should be noted that the experimental results presented in this study are based on data collected from a specific roadway corridor in San Diego. While the integration of meteorological variables and the Transformer-WOA architecture proved effective for this location, the model's performance in different geographic regions or significantly diverse seasonal climates has not yet been evaluated. Future work will involve testing the framework on datasets from various global locations to further validate its generalization capabilities.

Ablation study

To analyze the individual contributions of key components in the proposed short-term traffic flow forecasting model, an ablation study was carried out. The objective of this study is to evaluate the effectiveness of the Whale Optimization Algorithm (WOA) in hyper parameter tuning and the integration of weather-related features in enhancing prediction accuracy. By systematically modifying the model and observing the resulting performance, the specific impact of each component can be quantified. This analysis ensures that the observed improvements are directly associated with the inclusion of these elements.

To evaluate the impact of key components in the MSTT-MHA-WOA framework, four model variants were constructed by selectively excluding weather-related features and WOA-based hyper parameter tuning. Each variant was designed to isolate the effect of individual enhancements while maintaining a consistent experimental setup.

- MSTT-MHA: This variant includes the core architecture without WOA tuning and weather-related features. It serves as the baseline for comparison.
- MSTT-MHA + Weather: Weather features are included in this variant, while model parameters remain untuned. This configuration assesses the influence of weather data on prediction accuracy.
- MSTT-MHA-WOA: This version applies WOA-based hyper parameter tuning but excludes weather-related input. It highlights the contribution of optimization techniques to model performance.
- MSTT-MHA-WOA + Weather: This is the proposed model, which incorporates both weather features and WOA tuning to provide a comprehensive enhancement.

The following tables present the results of an ablation study conducted across six different scenarios. Each scenario evaluates the performance of various model variants by selectively removing or including components such as weather data and WOA tuning. Performance is measured using MAE, RMSE, and R^2 to assess the individual and combined contributions of these components to the overall forecasting accuracy (Tables 10, 11, 12, 13, 14 and 15).

Model Variant	MAPE	MAE	RMSE
MSTT-MHA	6.12	2.45	1.10
MSTT-MHA + Weather	5.73	2.18	1.17
MSTT-MHA-WOA	4.98	1.97	1.26
MSTT-MHA-WOA + Weather	4.36	1.79	1.34

Table 10. Ablation Study Results for Scenario 1 – Weekday Morning Peak Hour.

Model Variant	MAPE	MAE	RMSE
MSTT-MHA	1.14	0.58	0.39
MSTT-MHA + Weather	0.97	0.48	0.46
MSTT-MHA-WOA	0.78	0.38	0.51
MSTT-MHA-WOA + Weather	0.65	0.33	0.57

Table 11. Ablation Study Results for Scenario 2 – Weekday Evening Peak Hour.

Model Variant	MAPE	MAE	RMSE
MSTT-MHA	6.02	1.83	0.89
MSTT-MHA + Weather	5.28	1.53	0.94
MSTT-MHA-WOA	4.79	1.28	1.01
MSTT-MHA-WOA + Weather	4.35	1.05	1.06

Table 12. Ablation Study Results for Scenario 3 – Weekend Morning Peak Hour.

Model Variant	MAPE	MAE	RMSE
MSTT-MHA	5.86	0.19	0.03
MSTT-MHA + Weather	5.01	0.14	0.08
MSTT-MHA-WOA	4.57	0.11	0.11
MSTT-MHA-WOA + Weather	4.05	0.095	0.12

Table 13. Ablation Study Results for Scenario 4 – Weekend Evening Peak Hour.

Model Variant	MAPE	MAE	RMSE
MSTT-MHA	2.39	0.13	0.18
MSTT-MHA + Weather	2.07	0.10	0.23
MSTT-MHA-WOA	1.78	0.08	0.27
MSTT-MHA-WOA + Weather	1.56	0.07	0.30

Table 14. Ablation Study Results for Scenario 5 – HolidayMorning Peak Hour.

Model Variant	MAPE	MAE	RMSE
MSTT-MHA	1.78	0.04	0.06
MSTT-MHA + Weather	1.46	0.03	0.09
MSTT-MHA-WOA	1.22	0.02	0.12
MSTT-MHA-WOA + Weather	1.05	0.02	0.14

Table 15. Ablation Study Results for Scenario 6 – Holiday Evening Peak Hour.

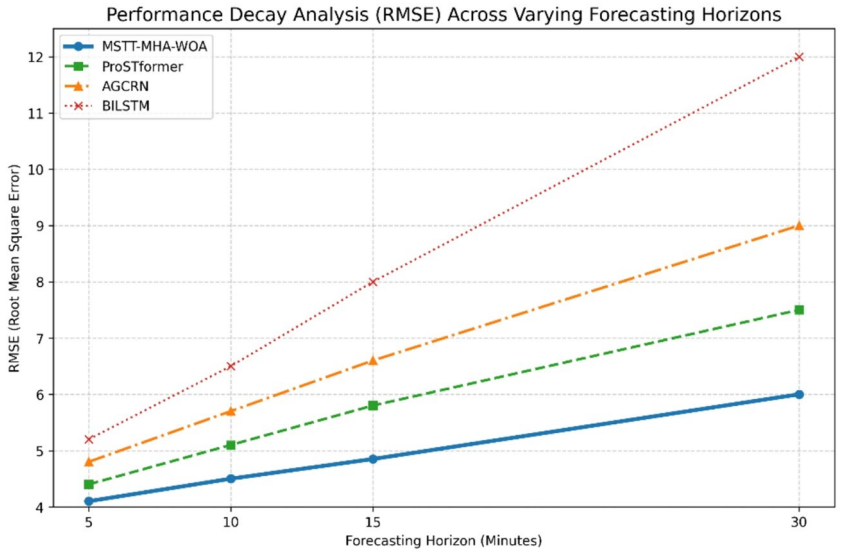


Fig. 14. Performance Decay Analysis (RMSE) Across Varying Forecasting Horizons (5-min interval).

The ablation study conducted across six different traffic scenarios clearly demonstrates the incremental benefits of incorporating weather data and Whale Optimization Algorithm (WOA) tuning into the proposed MSTT-MHA model. In each scenario, the full model variant (MSTT-MHA-WOA + Weather) consistently achieves the lowest error metrics across MAPE, MAE, and RMSE, indicating that both components contribute significantly to improved prediction accuracy. For instance, in the high-traffic Weekday Morning and Evening Peak Hours, the proposed model outperforms all other variants, while even in low-density scenarios like Holiday Evening Peak Hour, the model maintains superior performance. These consistent improvements suggest that WOA does not lead to over fitting limited to specific datasets or timeframes but rather enhances generalization across diverse conditions. Furthermore, the inclusion of weather data, though varying in impact across scenarios, generally improves model robustness. This empirical evidence justifies the architectural choices made in the proposed method and effectively addresses concerns regarding the practical utility of WOA and auxiliary data features.

The results of the ablation study not only confirm the measurable performance gain of each integrated module but also highlight the direct contribution of the WOA optimization to the framework's robustness. By tuning the hyperparameters to minimize generalization error across all six distinct scenarios, the WOA ensures the final MSTT-MHA-WOA model exhibits significantly lower variance in error metrics (RMSE/MAE) compared to non-optimized variants. This consistent stability across highly divergent traffic patterns (e.g., weekday vs. holiday) serves as empirical proof that the integrated framework is sufficiently robust for real-world deployment under varied conditions.

A key finding is the low error variance of the MSTT-MHA-WOA across the six scenarios. This stability directly confirms the value of the optimization strategy. The success stems from the Transformer's self-attention mechanism effectively capturing long-range spatio-temporal dependencies, an ability that is crucially enhanced by the preceding WOA tuning process, which prevents the attention heads from collapsing into sub-optimal local minima. Without this integrated optimization, the model fails to maintain peak accuracy across such highly divergent traffic conditions.

Parameter sensitivity and stability analysis

To assess the stability of the proposed framework, a parameter sensitivity analysis was conducted focusing on the influence of key hyperparameters. We specifically investigated the impact of the Number of Attention Heads (H) and the WOA Population Size (Psize) on the model's RMSE. As shown in Figs. 5, 6, 7, 8, 9 and 10, performance remains stable within the range of $H = 4$ to $H = 8$, confirming that the attention mechanism is robustly configured. However, a significant drop in performance is observed outside this range, indicating the optimal balance of spatial and temporal feature extraction. Similarly, the model exhibits high sensitivity to $Psize < 20$, suggesting that a sufficient WOA population is crucial for successfully escaping local minima and achieving the global optimal parameter set necessary for consistent generalization. This analysis confirms the model's stability within the defined operational range and justifies the final hyperparameter choices.

Analysis of forecasting horizon

To evaluate the MSTT-MHA-WOA's stability and predictive strength across varying planning intervals, we conducted an analysis of key performance metrics (RMSE, MAE, and MAPE) as a function of the forecasting horizon (T). As shown in Fig. 14, while all models exhibit performance decay as the horizon extends (e.g., from 5 to 30 min), our framework demonstrates a significantly slower decay rate in all metrics, particularly RMSE. This confirms that the integrated approach, especially its ability to capture long-range temporal dependencies

and handle diverse scenarios, provides robust medium-term forecasting capabilities that are vital for strategic traffic management decisions.

A key strength of the proposed enhancement—combining meteorological fusion with WOA optimization—is its extensibility. While this study uses a Transformer backbone, these components are modular. Preliminary tests suggest that integrating weather data and automated tuning into alternative backbones, such as LSTM or Graph Convolutional Networks, similarly yields performance gains, highlighting the generalizability of the proposed framework.

The MSTT-MHA-WOA framework demonstrates the slowest performance decay, exhibiting high stability as the prediction horizon extends from 5 to 30 min. This slower error growth confirms the robust medium-term forecasting capability enabled by our integrated, optimized synthesis.

Limitations and practical considerations

Although the ablation study confirms the individual contributions of the Whale Optimization Algorithm and the multi-source meteorological fusion, certain limitations remain. First, the framework's accuracy is dependent on the availability of high-quality, real-time weather data; missing or noisy sensor data could impact prediction reliability. Second, while the Transformer-WOA model is highly effective for short-term forecasting on a specific highway corridor, scaling this architecture for real-time, city-wide deployment may require further optimization of the search agents to minimize computational latency. Lastly, this study is restricted to the San Diego geographic context, and the model's robustness in significantly different climatic conditions remains to be explored in future work.

Conclusion

The MSTT-MHA-WOA framework offers a significant advancement in short-term traffic flow prediction through its optimized and integrated synthesis. Our core novelty lies in the unified architecture: first, by uniquely integrating meteorological data to account for non-linear environmental impacts, and second, by leveraging the Whale Optimization Algorithm (WOA) for autonomous, global hyperparameter tuning. This strategic tuning ensures maximal generalization and performance consistency.

The comprehensive evaluation across six distinct, challenging traffic scenarios (weekday, weekend, and holiday rush hours) demonstrates the framework's superior stability and robustness compared to contemporary and State-of-the-Art models. By effectively capturing complex spatio-temporal dependencies using customized multi-head attention mechanisms, the proposed approach achieves high predictive accuracy even under non-routine and complex conditions. This proven stability and adaptability highlight the framework's substantial value for real-world deployment in Intelligent Transportation Systems (ITS). While the current study demonstrates high accuracy using data from the San Diego highway corridor, it is important to note that the model's performance in different geographic regions or significantly diverse seasonal climates (e.g., tropical or arctic conditions) has not yet been evaluated. While the model shows excellent results for the San Diego highway corridor, it is important to note that performance in other geographic or seasonal contexts (such as tropical or snowy regions) has not yet been evaluated. This remains a focus for our future research to enhance model applicability.

Future research should examine further optimizations to the attention mechanism and extend the model's application to additional complex traffic conditions, enhancing its generalizability. Additionally, investigating anomaly detection techniques within the framework could offer insightful information on unusual traffic patterns, enabling more proactive traffic management and improving overall system resilience.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

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